Date .....

SMIB

↓

SG → GFM + T.L + Infinite Bus

↓

$E_g \angle \delta_g$ behind X_g

→ Syn. Gen represented by a reduced one-axis model

→ It was Controlled Vol Source with fixed mag & angle

→ Old GFM model Without dynamic Controller

↓

$E_g \angle \delta_g \rightarrow$ Constant

↓

$u = [\Gamma \;\; \delta \;\; \Delta\omega \;\; E'_q \;\; E_{fd}]^T$

↓

Adding GFM Controller

↓

$\delta_g = \delta_g(t)$

→ We've four differential states of Syn Gens

↓

$\delta, \Delta\omega, E'_q, E_{fd}$

→ (9)

Date .....

Derived q- Equations initially:

→ Rotor angle = δ

$$\frac{d\delta}{dt} = \Delta\omega \quad -①$$

→ Swing Equation

$$\frac{d\Delta\omega}{dt} = \frac{\pi f_0}{H}(T_m - T_e)$$

→ Transient EMF equation

$$\dot{E}'_{qv} = \frac{E_{fd} - (x_d - x'_d)i_d - E'_q}{T'_{do}}$$

→ AVR equation

$$\dot{E}_{fd} = \frac{K_A(V_{ref} - V_t) - E_{fd}}{T_A}$$

→ electrical Torque

$$T_e = E'_q i_q + (x_q - x'_d) i_d i_q$$

→ q-axis Current

$$i_q = \frac{x_g E_b \sin\delta - x_e E_g \sin(\delta_g - \delta)}{x_q x_g + x_q x_e + x_g x_e}$$

D_q

→ d-axis Current

$$i_d = \frac{(x_g + x_e)E'_q - x_g E_b \cos\delta - x_e E_g \cos(\delta_g - \delta)}{x_g x_e + x_g x'_d + x_e x'_d}$$

D_d

$$V = V_t e^{j\theta}$$

$$V = V_d + jV_q$$

$$V_t e^{j\theta} = (V_d + jV_d)\, e^{(\delta - \pi/2)}$$

$$V_d = x_q i_q \quad , \quad V_q = E'_q - x'_d i_d$$

$$V_t = \sqrt{V_d^2 + V_q^2}$$

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→ GFM Active & Reactive power

↓

The GFM current is determined by the voltage difference b/w

$E_g \angle \delta_g$ & $V_t \angle \theta$ through X_g.

→ The complex power injected by the GFM is

$$S_g = P_g + jQ_g$$

The active pow: $P_g = \frac{V_t E_g}{X_g} \sin(\delta_g - \theta)$

& Reactive power $Q_g = \frac{V_t}{X_g} [E_g \cos(\delta_g - \theta) - V_t]$

↓

$\delta_g - \theta \longrightarrow P_g$

$E_g - V_t \longrightarrow Q_g$

⇒ Using ① Active power Synchronization.

↳ Set the power angle of the GFM power angle based on the difference b/w P^* & P-actual

$P_g^* - P_g \longrightarrow$ Controls frequency

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As GFM behaves as a Voltage source with a low Series impedance → uses a power syn loop without PLL

↓

A Sync loop sets the power angle 'δg' which determines the q-axis position of internal Voltage Source '$E_g e^{j\delta_g}$' with respect to ref Vol $V_{oo}\angle 0°$ based on the difference $P_g^* - P_g$

↓

As $P_g = (V_t/x_g) E_g \sin(\delta_g - \theta)$

↓

the active power depends on δ_g & Voltage at the output terminal '$V_t e^{j\theta}$'

the rotation speed of the dq-axis is denoted by ω_g "$\omega_g - \omega_s$" → p.4

power angle is calculated as

$$\frac{1}{\omega_s}\frac{d\delta_g}{dt} = \omega_g - 1$$

→ GFM droop equation is given as

$$\Delta\omega_g = \frac{\Delta P_g^* - \Delta P_g}{J_g s + D_g}$$

J_g → Inertia Constant of GFM

D_g → damping Constant

$R_g = \frac{1}{D_g}$ → Droop Constant

↓

ratio of normalized freq variation $\Delta f/f_n$ & normalized power deviation $\Delta P/P_n$.

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Droop sys → responds to active-power error
Washout → responds to power change while maintaining SS

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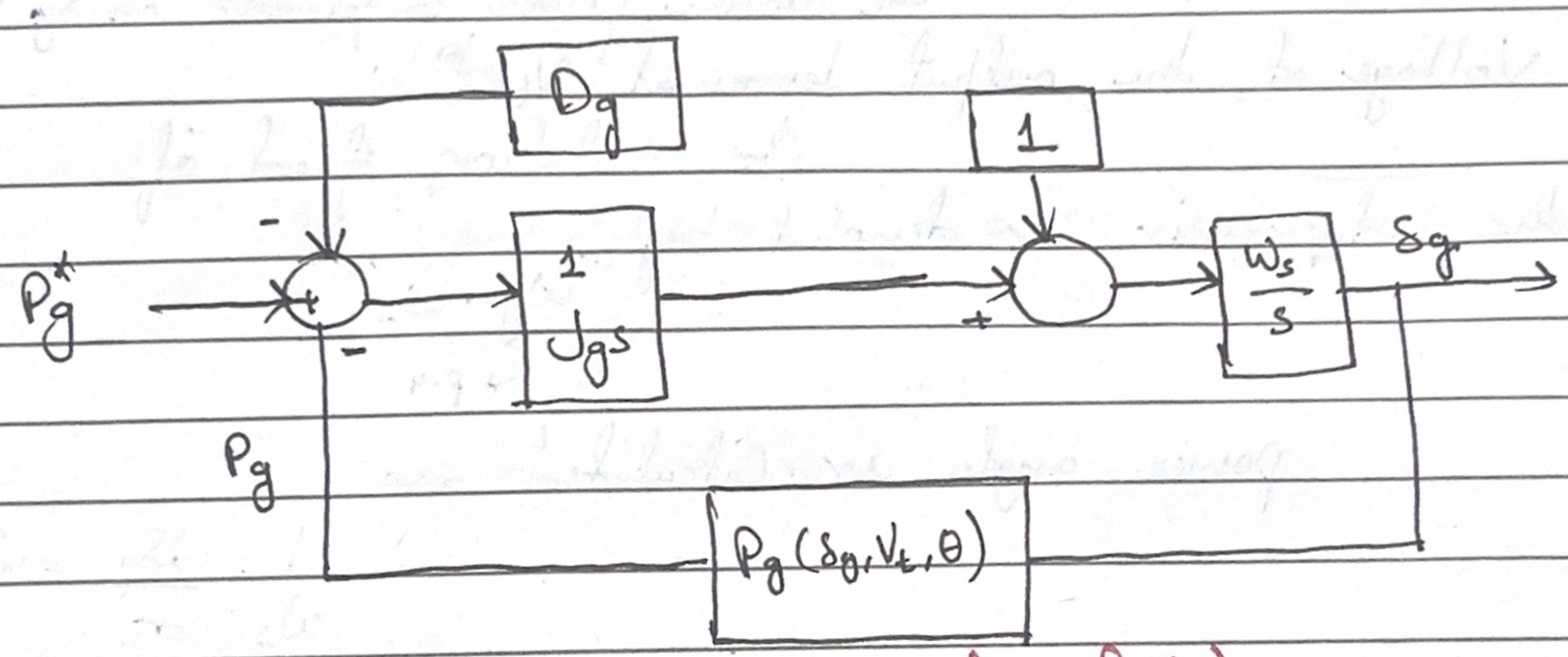
If Inertia constant is equal to Zero, ω_g is calculated as product of a droop constant & active power increment w.r.t ref.

ω_g

$\therefore J_s = 0$

$$\therefore \Delta\omega_g = \frac{1}{D_g}(\Delta P_g^* - \Delta P_g)$$

$$\Rightarrow \Delta\omega_g = R_g(\Delta P_g^* - \Delta P_g)$$



(GFM Power Synchronization loop)

→ Adding the POD-P Stabilizer

↓

Power oscillation damping using a GFC is carried out by acting on the power Synch. loop. By measuring the frequency at the output terminals of GFM, ω, freq ω in Syn loop is determined using a washout function

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In the POD-P washout function, the measured frequency deviation passes through a washout filter & modifies the GFM frequency

↳ which determine $\delta_g(t)$

→ acts as HPF to ensure low-freq oscillations r not affect

The washout function is

$$\omega' = K_w \frac{T_w s}{T_w s + 1}$$

To convert this transfer function into a state space equation, introduce 'z' as low pass filter state.

$$T_w \dot{z} = \Delta\omega_{pu} - z$$

↳ $\Delta\omega_{pu} = \frac{\Delta\omega}{\omega_s}$

$$\therefore \dot{z} = \frac{1}{T_w}\left(\frac{\Delta\omega}{\omega_s} - z\right)$$

The washout output is

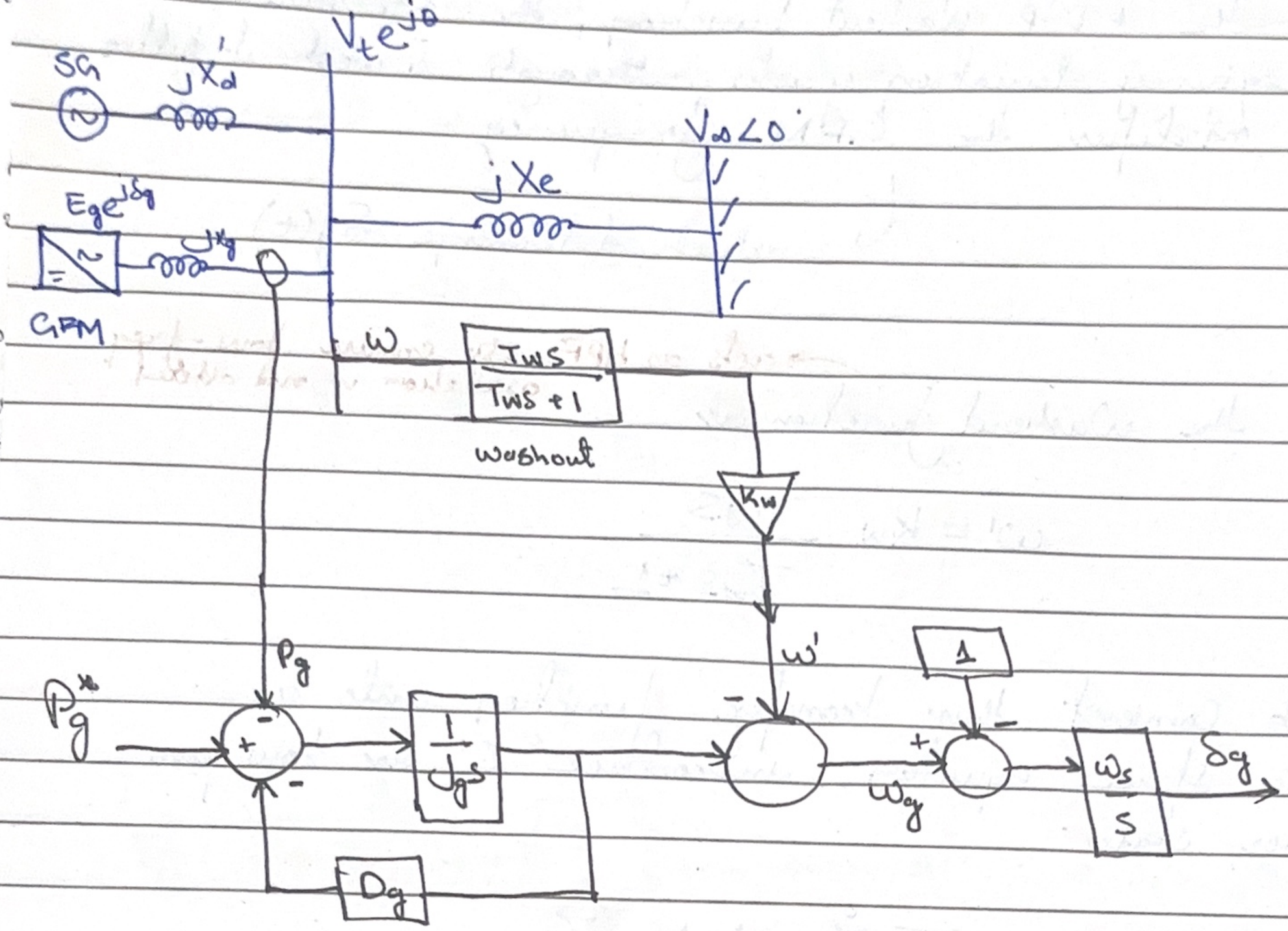
$$\Delta\omega_{poo} = K_w\left(\frac{\Delta\omega}{\omega_s} - z\right)$$

if $P_g < P_g^*$

$P_g^* - P_g > 0$

$\therefore \omega_g > 1$

so Datfreq ↑.....



→ the droop contribution is $R_g(P_g^* - P_g)$

the POD contribution is $-K_w\left(\frac{\Delta\omega}{\omega_s} - z\right)$

$\therefore \quad \Delta\omega_g = R_g(P_g^* - P_g) - K_w\left(\frac{\Delta\omega}{\omega_s} - z\right)$

Droop contribut → POD-P Stabilizer

Thus

$$\omega_g = 1 + R_g(P_g^* - P_g) - K_w\left(\frac{\Delta\omega}{\omega_s} - z\right)$$

↓

Combining Droop + Washout (POD-P)

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Here,

↓

the proposed control system acts on angle δ_g keeping Voltage magnitude E_g constant.

↓

If E_g matches V_t, $\Delta\omega \approx 0$

→ Deriving the dynamic 'δ_g' equation

↓ We know $\dot{\delta}_g = \omega_s(\omega_{g-1})$

$$\omega_{g-1} = R_g(P_g^* - P_g) - k_w\left(\frac{\Delta\omega}{\omega_s} - z\right)$$

$$\therefore \dot{\delta}_g = \omega_s\left[R_g(P_g^* - P_g) - k_w\left(\frac{\Delta\omega}{\omega_s} - z\right)\right]$$

↓

5th diff eqn/state.

$$\dot{\delta}_g = \omega_s R_g(P_g^* - P_g) - \omega_s k_w \frac{\Delta\omega}{\omega_s} + \omega_s k_w z$$

Since $\frac{\omega_s \Delta\omega}{\omega_s} = \Delta\omega$

$$\therefore \dot{\delta}_g = \omega_s R_g(P_g^* - P_g) - k_w\Delta\omega + \omega_s k_w z$$

↓

Now Connecting P_g back to electrical system

As $P_g = \frac{V_t E_g}{X_g}\sin(\delta_g - \theta)$

$$\therefore \boxed{\dot{\delta}_g = \omega_s R_g\left[P_g^* - \frac{V_t E_g}{X_g}\sin(\delta_g - \theta)\right] - k_w\Delta\omega + \omega_s k_w z}$$

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Here, 'θ' is not independent dynamic state, rather its angle of generator bus.

$$V_d = X_q I_q$$

$$V_q = E'_q - X'_d I_d$$

$$V_t = \sqrt{V_d^2 + V_q^2}$$

$$\theta = \tan^{-1}(V_q / V_d) + \delta - \frac{\pi}{2}$$

Calculated from electrical o/p

$$I_q = \frac{X_g E_b \sin\delta - X_e E_g \sin(\delta_g - \delta)}{X_q X_g + X_q X_e + X_g X_e}$$

$$I_d = \frac{(X_g + X_e) E'_q - X_g E_b \cos\delta - X_e E_g \cos(\delta_g - \delta)}{X_g X_e + X_g X'_d + X_e X'_d}$$

Now Non-linear model becomes

$$x = [\delta \;\; \Delta\omega \;\; E'_q \;\; E_{fd} \;\; \delta_g \;\; z]$$

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Date .....

→ rotor angle

$$\dot{\delta} = \Delta w \quad \text{—①}$$

→ Swing

$$\dot{\Delta w} = \frac{\pi f_0}{H}(T_m - T_e) \quad \text{—②}$$

→ EMF

$$\dot{E}'_{qv} = \frac{E_{fd} - E'_v - (X_d - X'_d) I_d}{T'_{do}} \quad \text{—③}$$

→ AVR

$$\dot{E}_{fd} = \frac{K_A(V_{ref} - V_t) - E_{fd}}{T_A} \quad \text{—④}$$

→ GFM angle

$$\dot{\delta}_g = \omega_s R_g (P_g^* - P_g) - K_w \Delta w + \omega_s K_w Z \quad \text{—⑤}$$

→ Washout state

$$\dot{Z} = \frac{1}{T_w}\left(\frac{\Delta w - z}{\omega_s}\right) \quad \text{—⑥}$$

Spiral